THE PROBLEM

After triage, nobody is watching the waiting room.

A patient can look stable at triage, sit down to wait, and get worse for hours with no one positioned to see it.

673

deaths in Nova Scotia emergency departments in 2023-24. It was **406** in 2020-21.*

55,000

people left without being seen in 2022-23, about **10%** of 530,000 emergency visits.

32.8%

of emergency patients in an Edinburgh study got worse in at least one vital sign **within 4 hours**.

*Counts all deaths after triage and before discharge or admission, not only waiting-room deaths. Nova Scotia Health also cites sicker patients. Sources: Nova Scotia Health figures reported by CTV News and The Globe and Mail, July 2024. Reed et al., SNAP40-ED study (250 patients), Emergency Care Journal, 2021.

THE SOLUTION

ALLEY keeps watching while you wait.

It tells the nurse who needs help first. **It does not replace the nurse.**

1. **Camera: posture, seat by seat**
   Upright, slumped, or lying down. Computer vision (OpenCV, pose estimation). No facial recognition. No video stored.

2. **Wearable band: heart rate, blood oxygen**
   The kind of device hospitals are already piloting.

3. **Health record: triage, complaint, history**
   The same signal means more for a higher-risk patient.

→

WHAT THE NURSE SEES

- Watching
- Check soon
- Check now

Every flag spells out the reason.

When the patient is called in: a handoff timeline of everything that happened while they waited.

WORKING TODAY
Live camera posture detection, the rules brain, the nurse dashboard, the handoff summary.

SIMULATED TODAY
Wearable data (from a control panel) and patient records (mock data).

WHY IT WORKS

It cross-checks before it alarms.

CROSS-CHECKED

Fewer false alarms, by design

One mild signal alone: keep watching. A severe sign is flagged right away.

Two independent signals agree, weighed against the health record: **Check now.**

Example: **heart rate climbing and the camera sees a slump.**

EXPLAINABLE

Clear rules, not a black box

Every flag states its reason in plain words.

AI writes the handoff summary only. It never decides who is seen first.

THE HARDWARE IS REAL

Already piloted and studied

> **Dartmouth General, N.S.** Waiting patients wear a wireless pulse and oxygen monitor. Pilot started Jan 2024.

> **Royal Infirmary of Edinburgh.** A wearable caught deterioration a median **65 minutes** before manual checks.

> **Vancouver Coastal Health.** Study of two wearable monitors for waiting ER patients, 2025.

> **NHS England (London).** Open call for ER wearables, Feb 2026.

ALLEY is not a hardware bet. It is the brain and the camera that make it useful.

Sources: Nova Scotia Health, Sept 2024, and CBC News, May 2024 (Dartmouth). Reed et al., SNAP40-ED, Emergency Care Journal, 2021 (Edinburgh, 250 patients; 65 min is versus manual checks, n=31). VCH Research Institute, Sept 2025 (NEWS2 study). DigitalHealth.London with NHS England (London), Feb 2026.

MEASURABLE OUTCOME

More people caught in time.

MEASURE 1: EARLIER

Minutes gained

Time from ALLEY's flag to the next manual vitals check. Benchmark: **65 minutes** in the Edinburgh wearable study.

MEASURE 2: FEWER UNSEEN

Fewer people missed

Of the patients who got worse while waiting, **how many ALLEY flagged** before staff were aware.

MEASURE 3: NURSE TIME

False alarms

How many flags turned out to need **no action**. If nurses cannot trust it, it does not help.

NEXT STEP

A silent pilot in one Nova Scotia ER. ALLEY runs without sending alerts. Then we measure how early it would have flagged the patients who deteriorated.

We are not promising a number of lives saved. We are promising to measure.

Benchmark: Reed et al., SNAP40-ED, Emergency Care Journal, 2021. Median 65 minutes earlier than manual vital signs (n=31 of 250 patients). ALLEY has not yet been tested with patients.

ALLEY IS NAMED IN HONOUR OF

Allison Holthoff

Allison was 37. She died on December 31, 2022, after waiting hours in the emergency room at Cumberland Regional Health Care Centre in Amherst, Nova Scotia.

She was triaged at 11:14 a.m. and got worse while she waited. No one was positioned to see it.

ALLEY exists so that no one waits unwatched.